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Pyridin-2-yl-substituted smart polymers sensitive to thermally triggered side group cyclization

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ABSTRACT

In this study, new thermo- and pH-responsive polymers based on pyridin-2-yl monomers were synthesized using reversible addition–fragmentation chain-transfer polymerization. A UV–vis study of intramolecular thermal cyclization in the synthesized polymers confirmed that rapid and complete elimination of small molecules is possible in polymeric systems. This removal at high temperatures renders the process promising in the design of materials for drug delivery.

1. Introduction

Stimuli-responsive polymers are widely used in various biomedical applications. The importance of these materials is recognized especially in developing more efficient delivery systems for drugs [1] and genes [2], protein assembly mechanisms, [3] also therapeutic applications [4] or biological sensing [5,6]. The most important characteristic of these polymers is that they can change their physical or chemical properties in response to external triggers such as temperature, pH [7,8] glucose level [9], among others [10]. Usually, smart polymers undergo reversible transitions and return to their original state after the stimulus is removed. This shape transition is primarily based on secondary forces such as electrostatic interactions and hydrogen bonding. Thermoresponsive poly(*N*-isopropylacrylamide—PNIPAM) is an example of such a reversible transition [11–13]. Responsiveness of smart polymers is an important characteristic in drug delivery systems in which polymers are used as carriers of biologically active compounds or prodrugs.

Another type of transition in response to external triggers is irreversible transformation, which is observed in, among others, intramolecular cyclization under the influence of external triggers such as electromagnetic radiation and temperature. Based on previous studies with thermolabile protecting groups [14] for hydroxyl [15], carboxyl [16], and phosphate functional groups [17], the present study focused on intramolecular thermal cyclization as a process driving the controlled release of active compounds.

2. Results and discussion

In this study, a preliminary method to combine two major types of responsiveness, thermosensitivity (nonreversible) and pH sensitivity (reversible), was presented to maintain control over the release process from polymeric moiety. Using the irreversible transition of thermoresponsive polymers, small ionic fragments can be removed from polymers. This transition can result in an ionic system in which the poly(acrylic acid) main chain of polymers is negatively charged. Polymers based on poly(acrylic acids) are pH responsive and accept protons at a low pH, exhibiting a rapid increase in swelling capabilities. This leads to a change of responsiveness within the polymeric system, and such systems can be designed using intramolecular thermal cyclization [18]. This approach offers several new possibilities in the synthesis of polymeric materials, due to controlled release, can be used in drug delivery applications and environmental purification. In this study, polymers were synthesized using reversibleaddition–fragmentation chain-transfer (RAFT) polymerization [19].

To investigate the intramolecular release process in polymers, pyridin-2-yl substituted monomers were designed and developed in this

Abbreviations: PNIPAM, poly(*N*-isopropylacrylamide); RAFT, reversible addition–fragmentation chain-transfer polymerization; MAMs, more-activated monomers; PMPyAEA, poly(2-(methyl(pyridin-2-yl)amino)ethyl acrylate) (**5a**); PBPyAEA, poly(2-(benzyl(pyridin-2-yl)amino)ethyl acrylate) (**5b**); PMPyAPA, poly(3-(methyl (pyridin-2-yl)amino)propyl acrylate) (**5c**); p-TsOH, p-toluenesulfonic acid monohydrate; DDMAT, 2-(dodecylthiocarbonothioylthio)-2-methylpropanoic acid; NMR, nuclear magnetic resonance; COSY, Correlated Spectroscopy; HSQC, Heteronuclear Single Quantum Coherence; HMBC, Heteronuclear Multiple Bond Correlation; SEC, Size-exclusion chromatography.

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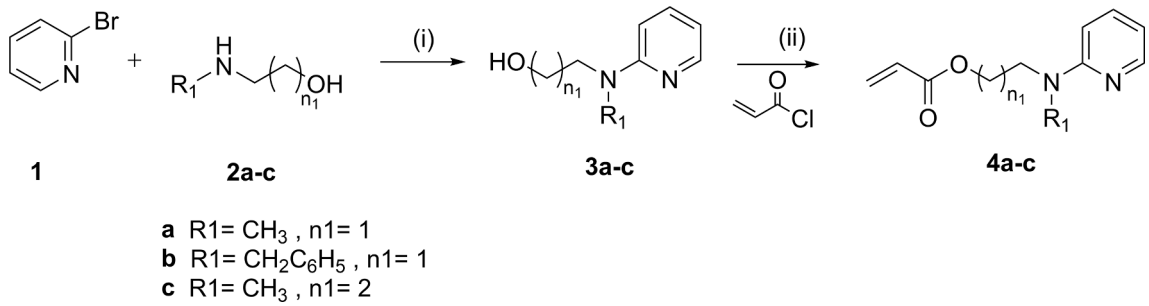
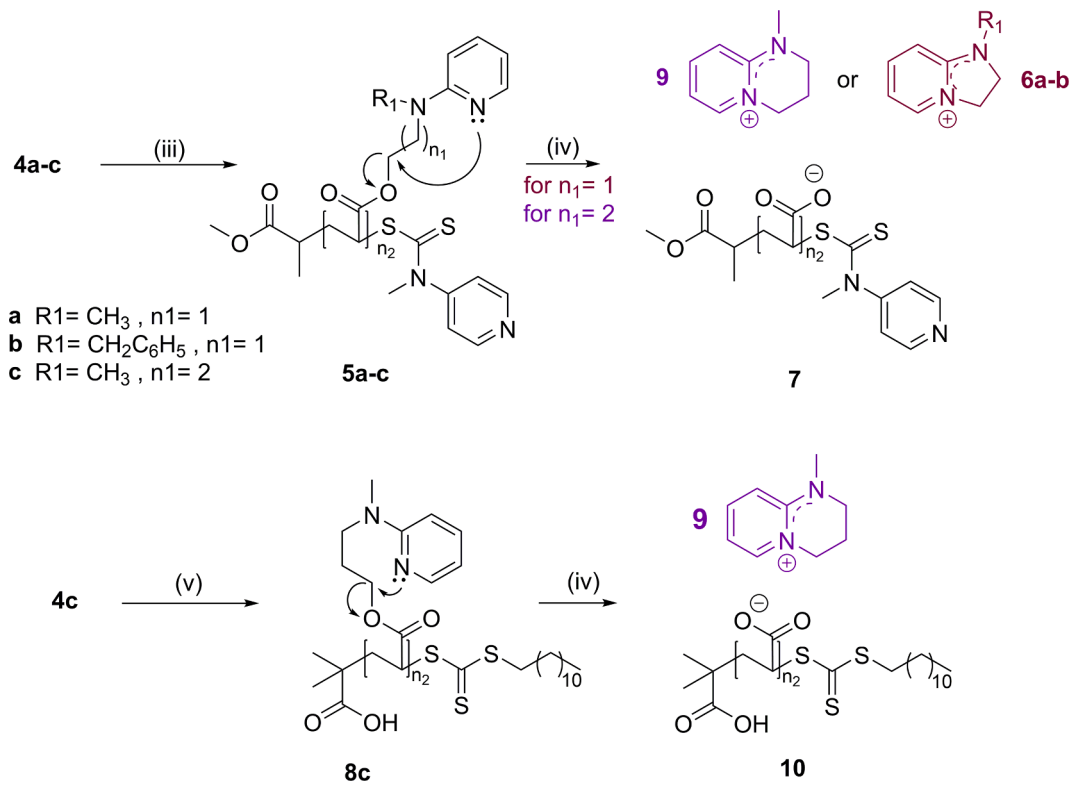


Scheme 1. Synthesis of monomers 4a-c. Reaction conditions: (i) triethylamine, DMF, 130 °C, microwave (MW), (ii) CH_2Cl_2, 0–25 °C.



Scheme 2. Synthesis of PMPyAEA (5a), PBPyAEA (5b), and PMPyAPA (5c) using a universal RAFT agent and PMPyAPA (8c) in non-acidic conditions, and the intramolecular thermal cyclization mechanism. Reaction conditions: (iii) universal switchable RAFT agent, ACHN, p-TsOH, 1,4-dioxane, N_2, 90 °C; (iv) 90 °C, H_2O or buffer solution; (v) DDMAT, ACHN, 1,4-dioxane, N_2, 90 °C.

study. For this purpose, the substituted amino alcohols (**3a**–**c**) were modified to obtain the acrylate monomers (**4a**–**c**) using reactions with acryloyl chloride (Scheme 1). Acrylate monomers, which are "more-activated monomers" (MAMs), were chosen for the polymerization technique to be used, i.e., RAFT polymerization. The monomers designed were polymerized to obtain three polymers whose structural modifications may impact the release process. The obtained thermoresponsive polymers are: poly(2-(methyl(pyridin-2-yl)amino)ethyl acrylate) (**5a**), poly(2-(benzyl(pyridin-2-yl)amino)ethyl acrylate) (**5b**), and poly(3-(methyl(pyridin-2-yl)amino)propyl acrylate) (PMPyAPA type—**5c** and **8c**).

Low-polydispersity, well-defined polymers were synthesized using the RAFT polymerization technique in which the universal switchable RAFT agent methyl 2-[methyl(4-pyridinyl)carbamothioylthio]propionate (Fig. S13, SI) was used in combination with 1,1′-azobis(cyclohexanecarbonitrile) as initiator (Scheme 2). Switchable chain transfer agents (CTAs) have to been protonated to control MAMs. [20,21] Low reactivity of C═S bond of the neutral form of universal RAFT agent result is faster fragmentation and better control of LAMs polymerization. On the other hand, protonation of the pyridine ring increase the reactivity of the C═S group, which facilitates the polymerization of MAMs. To accomplish this, acidic conditions are necessary due to the protonation of the nitrogen of pyridine. In addition, the pyridine nitrogen of monomers should be protonated to prevent potential intermolecular cyclization during polymerization (especially in the cases of **5a** and **5b**). In this study, the best conditions for both protonations: the activation of the RAFT agent and protection of the monomer were achieved using strong organic acids such as *p*-toluene sulphonic acid (*p*-TsOH, $pK_a = -2.8$). The less acidic dichloroacetic acid ($pK_a = 1.35$) was found to be unable to fully protect the monomer. Moreover, strong inorganic acids such as hydrochloric acid ($pK_a = -5.9$) affected the RAFT polymerization process. The amount of acid used in each polymerization process was calculated using the following equation:

$$[p-TsOH] = 1.5([M] + [RAFT]),$$

where [M] is the moles of the monomer and [RAFT] is the moles of the universal RAFT agent used in the reaction.

All polymerization reactions were carried out under standard RAFT

Table 1
Molar masses analysis for pirydyn-2-yl polymers.

Polymer	$M_{n,\ theory}$[c] ($gmol^{-1}$)	$M_{n,\ NMR}$[d] ($gmol^{-1}$)	Conv. (%)[e]	$M_{w,\ SEC}$[f] ($gmol^{-1}$)	$M_{n,\ SEC}$[f] ($gmol^{-1}$)	$Đ$[g]
PNIPAM[a]	6 000	3 300	74	5 600	4 800	1.16
PNIPAM[b]	11 600	11 100	86	13 100	11 300	1.16
5a	11 000	10 500	98	12 200	8 000	1.54
5b	11 600	13 000	96	9 000	6 000	1.49
5c	11 000	7 700	95	–	–	–
8c	11 600	6 000	95	20 900	10 800	1.94

[a] Polymerization with universal switchable RAFT agent.
[b] Polymerization with DDMAT.
[c] Theoretical molecular weight calculated from Equation 1(SI).
[d] Molecular weight determined by ^{1}H NMR analysis.
[e] Monomer conversion determined by ^{1}H NMR.
[f] Molecular weight determined by SEC (THF with 1%TBAB).
[g] $Đ$ (M_w/M_n) was calculated by SEC analysis.

polymerization conditions. Commercially available monomer (NIPAM) was used to produce PNIPAM as a reverence polymer, without the cyclization effect. The molar ratios of all compounds used in the polymerization reactions and the exact procedure are presented in Table S1, SI. To confirm that the protection of the monomers and the use of a switchable RAFT agent were necessary, attempts to polymerize monomers **4a**–**c** using different RAFT agents such as 2-(dodecylthiocarbonothioylthio)-2-methylpropanoic acid (DDMAT) were made, which does not require acidic conditions for polymerization. Polymers with monomers **4a** and **4b** (ethyl linker) could not be obtained using DDMAT as the RAFT agent since intramolecular thermal cyclization occurred faster than RAFT polymerization and the cyclization process competed with polymerization. Additionally, polymerization with DDMAT in acidic conditions gave a very low conversion of **4a-b** monomers, approx. 10 % after 10 h of reaction. However, the use of monomer **4c** (propyl linker) resulted in the synthesis of a polymer with no evident cyclization. PMPyAPA polymers with different end-groups in their structure are distinguished, **5a** is a polymer with switchable universal RAFT agent end-groups), and **8c** with DDMAT end-groups. The synthesis and thermorelease of these polymers were possible under standard conditions (Scheme 2). The structures of the polymers were confirmed using a variety of nuclear magnetic resonance (NMR) analyses. ^{1}H NMR spectra were obtained for **5a**–**c** and **8c** with broadened peaks typical of polymers, and chemical shifts and integrals were consistent with the proposed structures. To confirm the structures and ensure purity, detailed structural characterization of a representative polymer **5a** was carried out using 2D NMR analyses such as COSY, HSQC, and HMBC. Chromatographic analyses were carried out as described in SI using PNIPAM, which was synthesized as a reference polymer. Number average molecular weights from SEC measurements were compared with theoretical molecular weights (Equation S1, SI) and those obtained using NMR end-group analysis (Table 1). As reported in a previous study, the number average molecular weight of nondegradable synthetic polymers used as drug carriers needs to be below 50 kDa.[22] The targeted mass of the polymeric models investigated in this study was chosen accordingly, i.e.: approx. 11 kDa. A much lower number average molecular weight was chosen to stay below the renal threshold even after postpolymerization modifications. A quite good agreement was observed between the number average molecular weights of polymers **5a**–**c** determined using NMR and SEC, respectively. Polydispersity indices $\leq$ 1.5, typical of controlled radical polymerization reactions, were obtained for **5a**–**c** polymers. Molecular weights for **8c** deviated from the theoretical value under the applied analytical conditions and polydispersity is close to 2.0. We assume it is connected to the free carboxylic acid end-group, which forms a betaine structure with the basic pyrid-2-yl side groups which in turn affects SEC analysis. Also, in view of the high M_n-values determined by SEC, agglomerate formation cannot be ruled out, the more since the M_n determined by SEC is more than 3 times higher than the one determined by NMR.

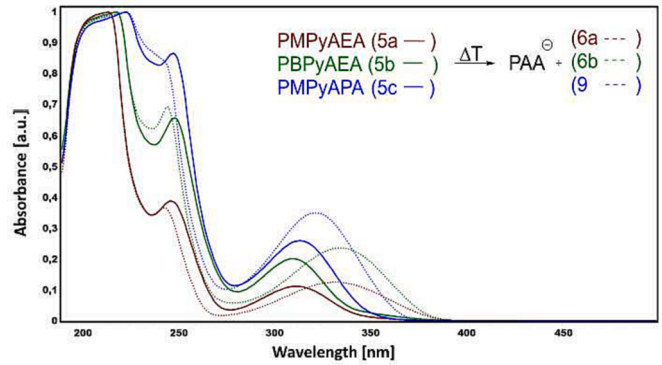


Fig. 1. UV–vis spectra of the intramolecular thermal cyclization process for thermo-responsive polymers. Solid lines show spectra for polymers before heating and dotted lines are for samples after Δt.

The most important objective of this study was to prove the controlled thermal release in model polymeric systems. Intramolecular thermal cyclization was previously investigated as the primary mechanism in thermolabile protecting groups [15]. Besides the protection function of compounds, the major focus was on the protected entity, i.e., nucleic acids. In the present study, deprotection as a release process was analyzed. During the cyclization process, a linear form of each repeating unit in a polymer (**5a**–**c**) was transferred into the cyclic form, resulting in an ionic system that contains a positively charged cyclic compound (**6a**, **6b** or **9**) and the negatively charged remaining poly(acrylic acid) main chain (**7** or **10**) (Scheme 2). The release process of the bicyclic compounds was evaluated using UV–vis profiles. A broad medium-intensity band at λ_{max} 310–314 nm was observed for all investigated polymers, which was attributable to the π–π* transition responsible for the thermoresponse of the polymers (Fig. 1). Then, aqueous polymer solutions were thermoset at 90 °C for several hours. Spectra were collected during this period, which allowed for observing the shifts of the bands over time. Due to the shorter time of deprotection, 90 °C was chosen to analyze the thermally triggered release process accurately. However, it is important to note that intramolecular thermal cyclization can also occur at a lower temperature more suitable for biological systems, what was already studied in our group for carboxylic derivatives of nucleosides, [16] though the time of release is mostly extended. For purpose of this research, a higher temperature was more suitable.

A red shift of a broad band was observed in UV–vis absorbance spectra for all investigated polymers (Fig. 1). These bathochromic shifts confirm that a transition from the linear to the cyclic form is possible in all polymers. Although intramolecular thermal cyclization is driven by the nucleophilic characteristic of the pyridinyl nitrogen, [23] other structural parameters can also control its rate. The change in the absorbance corresponds to the release process. As expected, the time (Δt) required to fully release the cyclic compound is dependent on the functional group at the amine nitrogen and the length of the alkyl linker. Shifts of 21 nm and 24 nm were observed for **5a** and **5b**, respectively. These shifts are attributable to the formation of the five-membered heterocyclic compounds **6a** and **6b**, in which electrons were almost equally distributed over both rings, thus promoting π–π* transitions. The formation of the six-membered cyclic compound **9** from the solution of **5c** (Fig. S34, SI) also resulted in a shift toward the red region; however, this shift was much less pronounced i. e. approximately 8–10 nm. In addition, the time required to complete the formation was much longer: the complete release of compound **9** required 22 h. The longer time for the formation **9** can be explained by the higher energy demand for that process, which is associated with the energy level required for the formation of the six-membered heterocyclic compound. In contrast, the five-membered heterocyclic compound **6a** was formed within 150 min

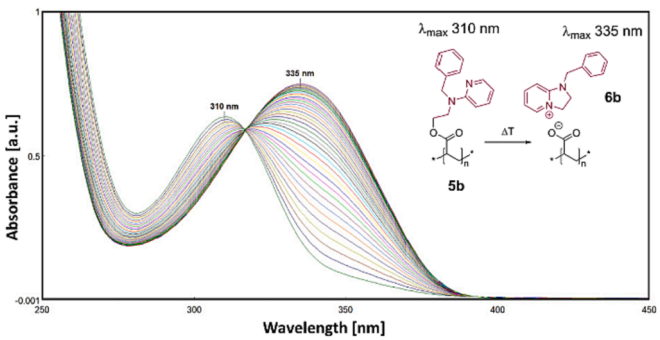


Fig. 2. Intramolecular thermal cyclization for PBPyAEA (**5b**), which is fully finished after Δt_b = 195 min (3 h 25 min). There were no changes in the spectrum after that time.

(Fig. S33, SI). A similar time was observed for the formation of compound **6b**, with a Δt value of 190 min. A similar change in absorption profiles was observed for methyl-substituted polymers (**5a** and **5c**), despite a wide disparity in Δt. A nonlinear absorbance change was observed during the formation of the benzyl-substituted bicyclic compound **6b** (Fig. 2). The mechanism of the formation of **6b** was different from that of methyl-substituted polymers. An isosbestic point was observed at 317 nm on the spectrum, which was not detected for other investigated polymers. A stable isosbestic point confirms that the reaction most likely proceeds without forming an intermediate. Furthermore, benzyl substituents result in higher electron delocalization, and conjugated systems achieve lower transition energies. As a result of intramolecular thermal cyclization, the absorbance peak for polymer **5b** moved to a longer wavelength (335 nm) than for other polymers. Noncovalent interactions such as π–π stacking that is theoretically possible for **5b** could be another important feature for drug delivery in polymer-based delivery systems. These interactions can result in additional stabilization between polymeric chains and increase the retention of aromatic drugs, such as DOX and paclitaxel, from polymeric micelles [24].

3. Conclusion

From a quantitative perspective, this study presented novel properties of stimuli-responsive polymers. Intramolecular thermal cyclization accounts for the transition from a thermoresponsive polymer to a pH-responsive one. We presumed that the proposed polymeric model in its final form, i.e., after the transition, can exchange and associate also positive ions such as copper. This can be an important characteristic in controlling protein folding in copper-sensitive bioconjugates. Future research should focus on copper ion binding of thermosensitive polymers to confirm the change of properties of the polymers-protein complex after thermorelease was confirmed. To enhance or decrease the time of formation of cyclic compounds, monomers can be modified in the desired manner, including linker elongation. The cyclization process can be inhibited, if necessary, using monomer protonation, which in turn allows for performing radical polymerization. The primary aim of the development of thermo-release polymers is to provide a tool for releasing a wide range of tailored molecules. In addition, the possibility to modify the cyclization process such that the bicyclic ring remains attached to the polymer while other (active) molecules are released is under research. In summary, the findings of this study showed that intramolecular thermal cyclization can indeed lead to a controlled release of small molecules from polymeric carriers and also that the time required for the release can be tailored.

Author contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: [Marcin Chmielewski reports financial support was provided by Institute of Bioorganic Chemistry Polish Academy of Sciences. Magdalena Paluch reports financial support was provided by National Center for Research and Development, Poland. Marcin Chmielewski reports a relationship with Institute of Bioorganic Chemistry Polish Academy of Sciences that includes: employment.].

Data availability

No data was used for the research described in the article.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2023.111865.

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